Roy (Yuanxi) Cheng

Laguna Niguel, CA | (949) 378-0596 | roycheng27@ucla.edu | linkedin.com/in/ycheng05

Education

University of California, Los Angeles (UCLA) Sep 2023 – Mar 2027
Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering *GPA: 3.75*

- **Awards & Programs:** Dean's Honor's List, Persolv AI Intensive, BCG Case Comp Semi-finalist
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Neural Networks & Deep Learning, Data Analysis & Regression, Data Mining, Probability & Statistical Inference, Databases, NLP, Monte Carlo, Optimization

Work and Leadership Experience

Capital One McLean, Virginia
Product Management and Analysis Intern *Jun 2026 – Present*

- Analyze corporate workforce data with SQL and Python, identifying effects and factors tied to successful org transitions
- Synthesize insights into Tableau dashboard of key reorg metrics, presenting to SVP to help HRBPs make reorg decisions
- Architect a 6-stage Python pipeline ingesting 1,459 Snowflake performance reviews through PII anonymization, BGE semantic filtering, and FAISS retrieval before routing to an LLM DWA scoring agent
- Build a custom LLM client with sliding-window token rate limiting and retry-aware backoff to maintain reliable multi-hour inference throughput across the production review-processing pipeline

Data Science Union Los Angeles, California
External Vice President *Dec 2025 – Present*

- Led external partnerships, marketing, and professional development to expand consulting opportunities for members
- Managed 3 directors and aligned board priorities to deliver quarterly and annual cross-functional objectives
- Cultivated industry and alumni partnerships to secure client data science projects and expand career programming

Kohl's Los Angeles, California
Project Manager and Data Analyst *Jun 2025 – Mar 2026*

- Analyzed 8.5M+ logistics records with SQL and Python, resolving data quality to build a reliable shipment-level dataset
- Conducted EDA in Vertex AI Workbench, engineering supply-side shipment features that sharpened transit time forecasts
- Developed two-stage XGBoost forecasting model with time-aware validation, improving arrival prediction MAE by 83%

ACTUM Digital Prague, Czech Republic
Data Analyst and AI Intern *Jun 2025 – Aug 2025*

- Spearheaded project tasks in Jira, presenting structured updates to manager that accelerated delivery timelines by 20%
- Integrated an MCP server and authored prompt workflows for an internal AI assistant, co-presenting the live demo to ACTUM executives
- Extracted structured data from 600+ PDFs with Python, consolidating results into a centralized reporting dataset

Projects and Research

Biomedical Tissue Imaging | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article evaluating CNN-based instance segmentation architectures with AP metrics, benchmarking cytoplasm boundary precision for biomedical imaging
- Converted 4k+ CytoNuke annotations to COCO format, standardizing training across 3 segmentation architectures
- Trained YOLOv8 instance segmentation models in PyTorch, improving cytoplasm AP50 by 5% over Cyto-RCNN baseline

COVID Pandemic Effects on Secondary Education | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned 100k+ student records across multi-level datasets with pandas, enabling accurate longitudinal analysis in Tableau
- Aggregated 4 years of test scores into time-series visualizations, revealing student performance impacts tied to pandemic
- Created interactive Tableau map linking geographic data to score changes across U.S. states for a public research website

Adrenaline and Cognitive Fatigue Effects on Athletic Performance | *R, ggplot2, tidyverse, dplyr*

- Designed 2^3 factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at $\alpha = 0.05$
- Validated ANOVA assumptions using log transforms and ggplot2 diagnostics, confirming normality and homoscedasticity

Technical Skills

Languages: JavaScript, Python, C++, Java, SQL, R
Libraries/Frameworks: pandas, NumPy, PyTorch, scikit-learn, Matplotlib, TensorFlow, Keras, XGBoost, LangChain, MLflow
Tools & Platforms: Git, GitHub, AWS, GCP, Claude Code, BigQuery, Snowflake, Vertex AI, Tableau, Jupyter Notebooks